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1.3.1 Addition and Subtraction of Decimal Numbers

Iniya has purchased notebooks for ₹ 46.50 and a pencil box for ₹ 16.50. How much she will get as balance if she paid ₹ 100 to the shop keeper?

Price of a note book = ₹ 46.50 ; Price of a pencil box= ₹ 16.50

To find the amount to be paid, we have to add the price of both the items.

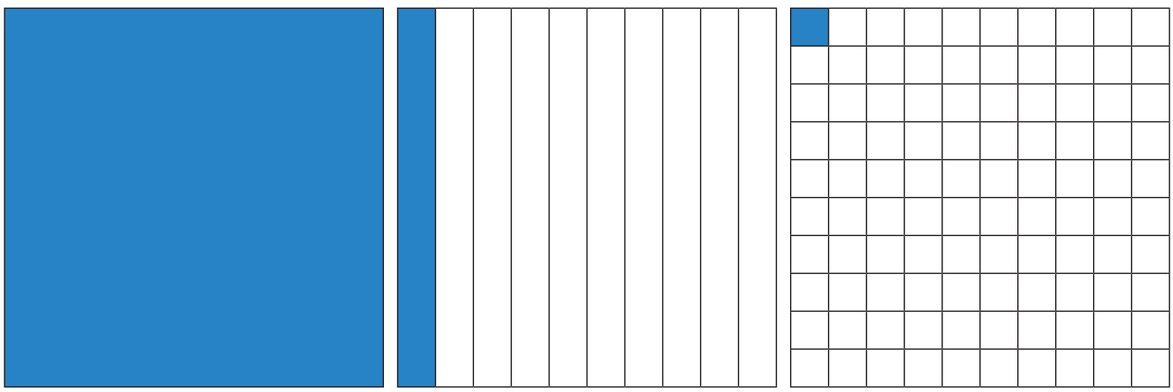
To get the balance amount we have to subtract the total expense from ₹ 100. To know the total expenses and the balance money, we need to understand addition and subtraction of decimals.

Addition and subtraction of decimals through models

Decimal grid or area models can be used to understand the process of addition and subtraction using decimal numbers.

(i) Grid model

We see below the grids to represent the decimal numbers 1.0, 0.1 and 0.01.



One whole, 10 by 10 Grid represents 1 or 1.0

Each column represents one tenth or 0.1

Each square represents one hundredth or 0.01

Having these grids let us try to do addition and subtraction of decimal numbers.

Example 1.5 Find the sum of 0.16 and 0.77 using decimal grid models.

Solution

Here, $0.16 = \frac{16}{100}$ and $0.77 = \frac{77}{100}$

First shade the region 0.16 and then shade 0.77.

The total shaded area is the sum.

So, 0.16 + 0.77 = 0.93 .

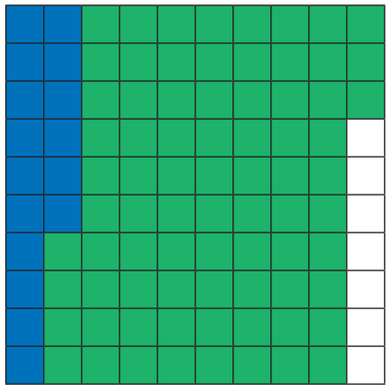


Fig. 1.2

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